



VoiceArmor-Edge

Real-Time AI Voice Scam Detection on Snapdragon-Powered HP PCs

Participant: Bharath G Devadiga (ECE Department)

College Email: bharath.23ec014@sode-edu.in

Target Hardware: Snapdragon X HP OmniBook + Arduino UNO Q

Competition: Snapdragon AI Lab Build & Present Challenge 2026

The Problem: AI Voice Scams & Impersonation

Scammers clone voices in seconds to trick people on live calls

- **Voice Cloning Scams:** Scammers use AI tools to clone voices of police officers, family members, or managers to steal money (such as recent Digital Arrest scams in India).
- **No Protection on PC:** Meeting apps like Microsoft Teams, Zoom, and WhatsApp cannot tell if an incoming voice is real or AI-generated.
- **Why Cloud Checks Fail:** Sending live call audio to a cloud server is too slow, needs constant internet, and leaks private conversations.
- **What We Need:** A fast, private tool running directly on the laptop that checks the audio locally and warns the user right away.

Why Use the Snapdragon Hexagon NPU?

Fast, private detection without slowing down your PC

- **Instant Warnings:** The Hexagon NPU checks voice patterns in under 15ms, warning you while the call is still happening.
- **100% Private:** Audio stays on your laptop. Zero data goes to the internet, and it works completely in Airplane Mode.
- **Smooth Laptop Performance:** The NPU handles the AI math in the background, leaving the CPU and GPU free for your video call.
- **Saves Battery:** The NPU runs efficiently at low power, so you can keep protection turned on all day without draining the battery.

How It Works: Audio Processing

Turning incoming sound into voice frequency pictures

- **Listen in Small Chunks:** Takes 250ms slices of incoming call sound from your speakers or headset.
- **Generate Spectrograms:** Converts raw sound waves into frequency pictures (called Mel-Spectrograms) showing sound energy over time.
- **Finding Fake Voice Clues:** AI voice cloning tools leave subtle artifacts in high frequencies that natural human voices do not have.
- **Fast & Lightweight:** Sound features are prepared quickly on the CPU and passed directly to the NPU.

The AI Model & Qualcomm AI Hub

Using lightweight, optimized models from Qualcomm's catalog

- **Proven Audio Model:** Built using compact audio classification models from the Qualcomm AI Hub library.
- **INT8 Quantization:** Converts model numbers to 8-bit integers so it runs fast on the Hexagon NPU and uses very little memory.
- **Qualcomm AI Hub Tools:** Compiled using Qualcomm's official tools specifically for Snapdragon X chips.
- **Direct NPU Format:** Exported as an ONNX model that runs directly on Qualcomm's hardware accelerator.

Software Setup on Windows

Clean, direct pipeline on Windows on Snapdragon

- **ONNX Runtime:** Runs using the official Qualcomm QNN execution plugin on Windows.
- **Hexagon Driver:** Connects directly to Qualcomm's QnnHtp.dll driver to use the NPU hardware.
- **Shared Memory:** Sound features move straight into NPU memory without unnecessary data copying.
- **Live Threat Score:** Updates the scam probability score every quarter of a second.

Adding the Arduino UNO Q

A physical warning light you cannot miss

- **Connected by USB:** The Snapdragon PC sends an alert byte to the Arduino board if a fake voice is detected.
- **Red Warning Light:** An external red LED and warning buzzer turn on immediately to grab your attention.
- **Tamper-Proof Alert:** Even if a scammer full-screens a window or covers your display, the physical light stays visible.
- **Safe Mode:** Shows a steady green light when normal human speech is verified.



Running on the HP OmniBook

Quiet, lightweight protection in the background

- **System Tray App:** Sits quietly in the Windows taskbar just like an antivirus tool.
- **Very Low Memory:** Uses under 120 MB of RAM and less than 3% CPU power.
- **No Complex Setup:** Automatically listens to your default call audio without changing app settings.
- **Works Offline:** Tested and fully working with Wi-Fi and Bluetooth turned off (Airplane Mode).



Meeting Competition Goals

Built directly to address Qualcomm's evaluation criteria

- **Technical Implementation:** Real audio signal processing, Qualcomm AI Hub model optimization, and Hexagon NPU offload.
- **Application & Innovation:** Solves real-world AI voice scams in India while keeping user data private.
- **Deployment on Snapdragon:** Designed specifically for Windows on Snapdragon HP PCs with low power draw.
- **Clear Documentation:** Backed by full code, clean setup instructions, and working scripts on GitHub.

Future Plans & Summary

Next steps to improve VoiceArmor-Edge

- **Indian Languages:** Train on more regional Indian accents and dialects for even higher accuracy.
- **Smart Call Detection:** Automatically wake up only when a call starts to save even more power.
- **Browser Plugin:** Show quick warning banners inside web meeting apps like Google Meet.
- **Summary:** VoiceArmor-Edge shows how Snapdragon PCs can protect everyday people from AI scams locally and privately.